Jayadithya R

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GENAI ENGINEER | LLM FOUNDATIONS, SYSTEMS & EVALUATION

GenAI Engineer shipping core AI infrastructure: custom Rust BPE tokenizers outperforming mBERT by 36%, distributed batch GPU control planes on Kaggle/AWS, and first-author IEEE research in VLM distillation (3.1x VRAM reduction at 92% quality).

SKILLS

AI/ML & LLMs: Retrieval-Augmented Generation (RAG) | LLM Tool Calling | LoRA Fine-Tuning (Unsloth) | Eval Harnesses | Hugging Face Hub | ChromaDB

Languages: Python | Rust | SQL | TypeScript

Backend & Cloud: FastAPI | Docker | AWS EC2 | PostgreSQL | SQLite | GitHub Actions | WebSockets

WORK EXPERIENCE

Blubridge Technologies, Chennai, TN | AI Research Intern | 07/2025 - 10/2025

- Implemented operator-fusion passes in MLIR for custom tensor dialects, reducing kernel launch overhead by 20% across benchmark workloads.
- Lowered high-level dialect representations to LLVM IR with dynamic tensor shape inference, eliminating manual bufferization bottlenecks.
- Engineered 40+ lit test suites covering pass correctness and IR round-trip stability, preventing compiler regressions in automated CI.

PROJECTS

tamil-tokenizer | *Rust, rayon, Aho-Corasick, Python, Hugging Face* (github.com/barryallen16/tamil-tokenizer)

- Designed 1,135-token conjunct base vocabulary in Rust and parallel Aho-Corasick batch encoder, beating mBERT by 36.7% and Llama-3 on held-out compression.
- Trained 30K and 120K-merge models on Indic corpora with CSR indexing, gating Hugging Face publishes on a 100% decode fidelity verification suite.

kaggle-automation | *Python, FastAPI, Docker, WebSockets, Telegram API* (github.com/barryallen16/kaggle-automation)

- Architected Dockerized FastAPI control plane sharding parameter sweeps across Kaggle GPU/TPU kernels, gated by 5-minute pre-flight validation checks to eliminate quota exhaustion across 220+ runs.
- Implemented 12-hour session boundary monitor with Telegram alerts and chunked artifact streaming directly to disk, avoiding memory exhaustion on multi-gigabyte zip files.

fitcheck | *React Native, Expo, Qwen3-VL, LoRA, PostgreSQL* (barryallen16.github.io/fitcheck-website)

- Conducted teacher-student distillation for first-author IEEE paper, generating 538K caption-QA pairs from 106K garment images with Qwen3-VL.
- Distilled model into an on-device student via LoRA, achieving 3.1x VRAM reduction at 92% teacher quality validated against human raters ($r=0.303$, $p<0.05$).

triagemate | *Python, Liquid AI d1-3B, PyTorch, LoRA, Hugging Face* (github.com/barryallen16/triagemate)

- Engineered CLI triage engine ranking Semgrep alerts by exploitability in 12-14 ms on Nvidia T4 using Liquid AI d1-3B, lifting AUC by +12% over 4B dense models on real CVEs.

EDUCATION

B.Tech in Artificial Intelligence and Data Science, Dr. M.G.R. Educational and Research Institute, Chennai, TN | 2026

GPA: 8.0/10.0 | First-Author IEEE TQCEBT 2026 Publication (DOI: 10.1109/TQCEBT67648.2026.11681836)

AWARDS, ACCOLADES AND CERTIFICATIONS

2026 | First-author paper: Multi-Modular AI Framework for Fashion Recommendation | IEEE TQCEBT 2026

2026 | Published 30K and 120K Indic tokenizers with 100% decode fidelity gate | Hugging Face Hub